HALITE

FULL FYP DEVELOPMENT TIMELINE

AI-Powered Maintenance Diagnostics and Asset Intelligence Platform

FYP I → FYP II | 2026–2027 | FAST-NUCES Islamabad

Development Objective

Build HALITE incrementally from a working diagnostic foundation into a complete maintenance intelligence SaaS platform.
The target is approximately **90–95% functional completion by FYP II Mid**, leaving the remaining period for validation, product refinement, Job Fair preparation, and final defense.

OCT 2026

20%

FYP I Mid — Working Diagnostic Foundation

- Maintenance-only product navigation
- Live asset history and knowledge graph
- Symptom-based diagnostic chat
- Evidence-linked diagnostic suggestions
- Existing conflict-detection capability
- Functional GitHub Actions CI/CD pipeline
- SRS, architecture, and team contribution evidence

Target: Working vertical slice, not 20% of every module.

FYP I FINAL

50%

Integrated Maintenance Intelligence MVP

- Full diagnostic session workflow
- Hybrid graph and semantic retrieval
- Fault ranking with explanations
- Record ingestion and extraction
- Conflict review and human adjudication
- Asset management and RBAC
- SaaS dashboard and session history
- Initial automated evaluation suite
- SDS and updated requirements traceability

FYP II EARLY

75%

Advanced Intelligence and Training

- Multi-turn diagnostic questioning
- Knowledge graph exploration
- Technician training scenarios
- Case-based assessment and feedback
- Recurring fault detection
- Fleet analytics and reporting
- Organization workspaces and audit logs
- Expanded diagnostic benchmark

FYP II MID

90–95%

Job-Fair-Ready SaaS Product

- All flagship workflows integrated
- Training and analytics fully demonstrable
- Diagnostic evaluation and ablation studies
- Usability testing and performance benchmarking
- Production-style deployment
- Product walkthrough and polished demo
- Security and reliability testing

FYP II FINAL

100%

Validation, Documentation and Defense

- Fix outstanding defects
- Finalize evaluation results
- Complete documentation and traceability
- Rehearse technical defense
- Prepare deployment and demonstration backup

Key Delivery Strategy

FYP I: Build and integrate the diagnostic intelligence core.
FYP II Early: Expand into training, analytics, and advanced reasoning.
FYP II Mid: Reach near-complete SaaS functionality.
FYP II Final: Validate, stabilize, document, and defend the product.

Note: These percentages are proposed milestone targets, not measured development progress. Exact FYP II dates will follow the official university academic calendar.